BE-205 · BTech First Year · UIT, Barkatullah University, Bhopal

Engineering Graphics & CAD — Short Notes

Unit-wise complete syllabus coverage · Hinglish notes · Prepared by Ajeet Soni

UNIT I Introduction, Scales, Conics & Special Curves

1. Engineering Drawing – Need & Classification

DEFINITION Engineering drawing ek **graphical language** hai jisme kisi object ka shape, size, material aur manufacturing detail lines, symbols aur notes se exact tarike se dikhaya jata hai. Isko "language of engineers" kehte hain – har country ka engineer same convention padh leta hai.

Need: idea ko accurately communicate karna, manufacturing / construction ke liye exact size dena, records maintain karna, design check karna aur ambiguity khatam karna (words se 3D shape describe karna mushkil hai).

Basis	Types
Nature / purpose	Artistic drawing (feelings, art) vs **Engineering drawing** (technical, exact)
Field	Machine drawing, Civil / building drawing, Electrical & electronic drawing, Structural, Production drawing
Method	Freehand sketch, Instrument drawing, Computer aided drawing (CAD)
Projection	Orthographic (multi-view) vs Pictorial (isometric, oblique, perspective)
Use in industry	Layout / assembly drawing, detail (part) drawing, working drawing, patent drawing

2. Drawing Instruments & Uses

Drawing board – soft wood board, sheet fix karne ke liye (D0–D3 sizes).

Mini-drafter – T-square + set-squares + protractor ka kaam ek saath; parallel lines kisi bhi angle par.

T-square – horizontal parallel lines; working edge par set-square slide hota hai.

Set-squares – 45° and 30°–60°; vertical / inclined lines (15° multiples).

Protractor – angles measure/mark karna.

Compass & divider – compass se circles/arcs; divider se distance transfer / line ko equal parts mein divide karna.

Bow compass / bow divider – chhote circles (dia < 25 mm) ke liye.

Scales – plastic/cardboard scales M1–M8 (e.g. M1 = 1:1 & 1:2.5).

French curves – irregular smooth curves (conics, cycloid) draw karne ke liye.

Pencils – grades: 2H/H (construction, thin lines), HB (lettering, outlines). H = hard, B = black/soft.

Others – eraser, erasing shield, drawing clips/pins, sand paper block, templates.

3. Indian Standards (BIS SP-46 : 2003) & Sheet Layout

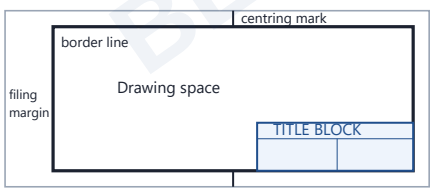
Bureau of Indian Standards ne **SP-46** "Engineering Drawing Practice for Schools & Colleges" diya hai (ISO based). Isme sheet size, lines, lettering, dimensioning, projection method (first angle preferred) sab standard kiye gaye hain.

Designation	Size (mm)	Border
A0	841 × 1189 (area 1 m^2)	20
A1	594 × 841	20
A2	420 × 594	10
A3	297 × 420	10
A4	210 × 297	10

Har next size = pichle ka half; sides ka ratio 1 : √2.

Layout of sheet:

- **Border lines** + left side par ~20 mm **filing margin**.
- **Title block** – bottom right corner (approx 185 × 65 mm): title, drawing no., scale, projection symbol, name, roll no., date, checked by.
- **Grid reference / zones**, **orientation & centring marks**, **trimming marks**.
- **Revision panel** and parts list (assembly drawing mein) title block ke upar.



Sheet layout (trimmed sheet, border, title block)

4. Conventions – Types of Lines

Line	Name	Use
A	Continuous thick	Visible outlines & edges
B	Continuous thin	Dimension, extension, leader, hatching lines
C	Thin freehand	Short break lines
D	Thin zig-zag	Long break lines
E / F	Dashed thick / thin	Hidden outlines & edges

Line	Name	Use
G	Thin chain (dash-dot)	Centre lines, axis, pitch circle, symmetry
H	Chain thin, thick at ends & turns	Cutting plane line
J	Chain thick	Surfaces with special treatment
K	Chain thin double-dashed	Adjacent parts, alternative positions of moving parts

Conventions mein materials ki hatching (steel, brass, wood, concrete), breaks of shafts/pipes, aur standard parts (springs, threads, gears) ke conventional symbols bhi aate hain. Thick : thin line width ratio ≈ 2 : 1 (e.g. 0.7 & 0.35 mm).

5. Technical Lettering

- Single-stroke capital letters, **vertical** ya **inclined (75° to horizontal)**.
- **Type A**: stroke thickness d = h/14 (thin, condensed). **Type B**: d = h/10 (most used in schools).
- Standard heights h: **2.5, 3.5, 5, 7, 10, 14, 20 mm**. Drawing no./title: 7–10 mm; sub-titles: 5–7 mm; dimensions/notes: 3.5–5 mm.
- Spacing between characters ≈ 2d (Type B: 0.2h), words ≈ 6d, lines ≈ 14d (baseline to baseline). Guide lines thin (2H) lagao.

6. Dimensioning

Dimensioning = drawing par size aur location ke numerical values + notes dena taaki part bina scale lagaye ban sake.

Elements: extension (projection) line – object se ~1 mm gap, dimension line se ~2–3 mm aage; **dimension line** (thin, arrowheads ke saath); **arrowhead** (length : width = 3 : 1, closed & filled); **leader line** (notes ke liye); dimension figure (value, mm mein, unit nahi likhte).

Types: size dimensions (length, dia) & location dimensions (centres, holes).

Systems: (1) **Aligned** – values dimension line ke upar, bottom ya right side se padhi jaayein. (2) **Unidirectional** – sab values horizontal, sirf bottom se padhi jaayein (dimension line beech se tooti).

Arrangement: chain (series), parallel, combined, progressive (from a datum) / co-ordinate.

> **EXAM TIP** Rules: har dimension sirf ek baar; hidden lines par dimension mat do; centre line / outline ko dimension line mat banao; dia ke liye Ø, radius ke liye R, square ke liye □; smaller dimension andar, larger bahar (lines cross na karein).

7. Scales

> **FORMULA**
>
> $$\text{R.F.} = \frac{\text{Length on drawing}}{\text{Actual length of object}} \ (\text{same units}) \ ; \quad \text{Length of scale} = \text{R.F.} \times \text{Max. length to be measured}$$

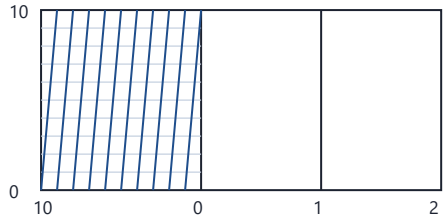
Types by RF: full size (1:1), reducing scale (1:2, 1:5, 1:10 ...; RF < 1), enlarging scale (2:1, 5:1 ...; RF > 1). Scale ka naam drawing par likhna zaroori.

(a) Plain scale – sirf 2 units (e.g. m & dm)

1. RF aur max length se scale length nikaalo; line draw karo (normally 150 mm ke aas-paas).
2. Usko main units mein divide karo; **zero** first division ke end par likho (right side numbering: 1, 2, 3...).
3. First division ko sub-units mein divide karo (left side numbering).
4. Scale ki width ~10 mm rakho, name, RF aur units likho; required length mark karke dikhao.

> **EXAMPLE** RF = 1/60, m & dm padhne hain, max 6 m. Scale length = 600 cm × 1/60 = **10 cm**. 10 cm ko 6 equal parts (each = 1 m) mein bato; first part ko 10 parts (each = 1 dm). 3.7 m dikhane ke liye: zero se right 3 main divisions + zero se left 7 sub-divisions.

(b) Diagonal scale – 3 units (e.g. m, dm, cm) – based on similar triangles



Diagonal scale – first main division mein 10 diagonals, 10 horizontal strips

Principle: ek short length *b* ko n equal parts mein divide karna ho to diagonal draw karo; k-th horizontal par diagonal ka horizontal offset = k·b/n. Isse 1/10 of smallest division read hota hai.

1. Scale length nikaalo, main units mein divide karo, 10 (ya n) parallel horizontal lines equal spacing par draw karo.
2. First main division ko top & bottom dono par 10 sub-divisions mein bato.
3. Bottom ke k-th point ko top ke (k+1)-th point se join karo (diagonals).
4. Reading = main + sub-division + (horizontal line no. × least count).

EXAMPLE RF = 1/4000, max 500 m, metres tak padhna hai. Scale length = 500 m × 1/4000 = 12.5 cm. 12.5 cm ko 5 parts (each = 100 m) mein bato; first part ko 10 parts (10 m) mein; 10 horizontal divisions se 1 m. 438 m dikhane ke liye: 4 main (400) + 3 sub (30) + 8th horizontal line par diagonal tak.

(c) Scale of chords – angles measure / set karne ke liye (protractor na ho to)

1. Line AO draw karo; O par perpendicular, radius R se quarter circle (quadrant) A se B tak banao.
2. Arc AB ko 9 equal parts (each 10°) mein divide karo (pehle 60° & 30° compass se, phir trisect).
3. A ko centre lekar har division point tak radius (chord) se arc kheecho jo AO extended line ko kaate – ye points 0°, 10°, 20° … 90° ki chord lengths hain.
4. Note: **chord of 60° = radius**; angle θ banane ke liye radius = chord of 60° se arc lo aur chord of θ cut karo.

8. Conic Sections

DEFINITION Right circular cone ko different planes se kaatne par mile curves = conics. **Eccentricity e = (distance of point from focus) / (distance from directrix)**. e < 1 → **ellipse** (plane cuts all generators), e = 1 → **parabola** (plane ∥ one generator), e > 1 → **hyperbola** (plane ∥ axis / cuts both nappes).

Curve	Construction methods
Ellipse	(1) **Concentric circle** method (2) **Oblong / rectangle** method (3) **Arcs of circles** (foci) method: PF_1+PF_2 = major axis (4) **Eccentricity** (directrix-focus) method (5) Parallelogram, loop-of-thread, trammel methods
Parabola	(1) Eccentricity (directrix-focus) method (2) **Rectangle** method (3) **Tangent** method (4) Parallelogram method
Hyperbola	(1) Eccentricity method (2) **Rectangular hyperbola** (asymptotes at 90°, PV = constant) (3) Foci / arcs method: $\lvert PF_1 - PF_2 \rvert$ = transverse axis

Ellipse by concentric circle method (most asked)

1. Major axis AB aur minor axis CD ko centre O par perpendicular draw karo; O se dono ke circles (dia = AB & CD) banao.
2. Dono circles ko 12 equal parts mein radial lines se divide karo.
3. Bade circle ke point se vertical line, chhote circle ke corresponding point se horizontal line – intersection = ellipse ka point.
4. Saare points ko French curve se smooth join karo. Foci: C se radius AB/2 se AB par cut → F_1, F_2.

Eccentricity method (any conic) – steps

1. Directrix DD aur axis (⊥ DD) draw karo; focus F distance par mark karo. Vertex V ko CF par is tarah divide karo ki VF/VC = e (e.g. e = 2/3 → CF ko 5 parts, V ko C se 3rd division par lo: VC = 3, VF = 2).
2. V par perpendicular VE = VF lo; CE join karke extend karo (ye line slope e deti hai).
3. Axis par points 1, 2, 3 … lo; har point par vertical line jo CE extended ko kaate (1′, 2′ …). Distance 1–1′ radius se F centre lekar vertical 1-1′ ko dono side cut karo → curve ke points.

Ellipse – oblong (rectangle) method: rectangle = major × minor axis. Half major axis OA aur rectangle ki side AE (= CD/2) dono ko same no. (e.g. 4) equal parts mein bato, dono ko A se number karo (1, 2, 3). C ko AE ke points se join karo; D (minor axis ka doosra end) se OA ke points se line extend karo; same number wali lines ke intersection = ellipse points (har quarter mein repeat).

Parabola – rectangle method: rectangle (base × axis height) banao. Half base ko n parts aur side ko n parts mein bato (base points axis se, side points vertex wale corner se number karo). Side ke points ko vertex V se join karo; base ke points se axis ke parallel lines – same number ke intersections = parabola points.
Hyperbola (rectangular): asymptotes OX, OY; given point P se lines ∥ OX, ∥ OY; O se radial lines – inse PV = const points milte hain.

TANGENT & NORMAL **Eccentricity curves:** point P ko F se join karo; F par PF ke ⊥ line draw karo jo directrix ko T par mile; **TP = tangent**, P par TP ke ⊥ = **normal**. **Ellipse (foci method):** PF_1, PF_2 join karo; $\angle F_1PF_2$ ka bisector = normal, uske ⊥ = tangent. **Parabola:** tangent, PF aur P se axis ke parallel line ke beech ke angle ko bisect karta hai.

9. Special (Roulette) Curves

Curve	Generation	Key data
Cycloid	Circle seedhi line par bina slip roll kare, circumference ke point ka locus	Base length = πD
Epicycloid	Generating circle (r) directing circle (R) ke **bahar** roll kare	Subtended angle θ = 360°·r/R
Hypocycloid	Generating circle directing circle ke **andar** roll kare	θ = 360°·r/R (if r = R/2 → straight line)
Involute	Taut string ko circle/polygon se unwind karne par end point ka locus	Circle: length = πD; gear teeth profile
Archimedean spiral	Point radius vector par uniform linear speed se move kare jab radius uniform angular speed se ghume	r = aθ (equal angle → equal increase in r); cams
Logarithmic (equiangular) spiral	Consecutive radius vectors (equal angles par) ka ratio constant	$r = a \cdot e^{b\theta}$, log r ∝ θ; tangent–radius angle constant

Cycloid – steps (circle dia D)

1. Circle draw karo, usse 12 equal parts mein divide karo; base line PA = πD draw karo aur usko bhi 12 parts mein bato.
2. Centre C se base ke parallel **locus of centre** line draw karo; uspe C_1, C_2 … C_{12} (base divisions ke upar) mark karo.
3. Circle ke division points se horizontal lines draw karo.
4. C_1 centre, radius D/2 se arc jo 1st horizontal line ko kaate → P_1; aise hi C_2 se 2nd line par P_2 … Smooth curve join karo.

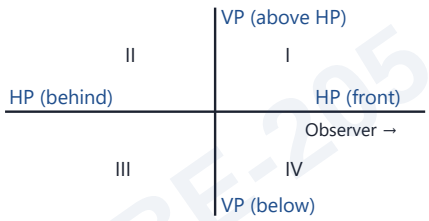
Epi/Hypo: same steps, bas base line ki jagah directing circle ka arc (angle θ) lo, 12 radial divisions, locus of centre ek arc (radius R + r ya R − r) hoga. **Involute of circle:** circle ko 12 parts mein bato, har point par tangent draw karo, tangent par length = (n/12)·πD mark karo → points join. **Archimedean spiral:** 12 radial lines aur radius ko 12 equal parts; circle 1 ka radial line 1 se intersection … point join.

NORMAL & TANGENT **Cycloid / epi / hypo:** point P se radius r leke locus of centre par C cut karo; C se base (ya directing circle radius) par perpendicular → N; **PN = normal**, ⊥ = tangent. **Involute:** P se base circle par tangent = normal. **Archimedean spiral:** constant C = $(r_1 - r_2)/(\theta_1 - \theta_2)$ (θ in radian); O se OP ke ⊥ OM = C lo; **MP = normal**.

UNIT II Projections, Points, Lines & Planes

1. Types of Projection

Type	Projectors	Result / use
Orthographic	Parallel to each other & ⊥ to plane of projection	2-D multi-view (FV, TV, SV) – true size; manufacturing drawing
Isometric (axonometric)	Parallel, ⊥ to plane; object tilted so 3 axes equally inclined (120°)	Single pictorial view, measurable along iso axes
Oblique	Parallel to each other but **inclined** to plane	Front face true shape; receding lines at 30°/45° (cavalier = full, cabinet = half length)
Perspective	**Converge** to station point (eye)	Realistic view, as eye sees; architecture. Size reduces with distance



Four quadrants (side view of HP & VP); observer right side se dekhta hai

Principal planes: VP (vertical, FV/elevation milta hai), HP (horizontal, TV/plan) aur PP (profile plane, side view). HP & VP ki intersection line = **reference line XY**. Drawing ke liye HP ko clockwise 90° ghuma kar VP ke saath ek plane mein laate hain.

Missing views: do given views (FV, TV) se teesra view (SV) banana.

1. Given views mein har corner ko number karo; object ko mentally visualise karo.
2. FV se horizontal projectors SV ki taraf; TV se horizontal projectors 45° **miter line** tak, wahan se vertical SV mein (depth transfer).
3. Intersections join karo; hidden edges dashed, centre lines chain.

First angle vs Third angle

First angle (BIS recommended)	Third angle (USA)
Object 1st quadrant mein: observer → object → plane	Object 3rd quadrant: observer → plane (transparent) → object
TV, FV ke **neeche**	TV, FV ke **upar**
LHSV, FV ke **right** mein	LHSV, FV ke **left** mein



Projection symbols (truncated cone) – title block mein lagate hain

2. Projection of Points

Quadrant	Position	FV (a')	TV (a)
I	Above HP, in front of VP	Above XY	Below XY
II	Above HP, behind VP	Above XY	Above XY
III	Below HP, behind VP	Below XY	Above XY
IV	Below HP, in front of VP	Below XY	Below XY

Rule: FV ki XY se distance = point ki HP se distance; TV ki XY se distance = VP se distance. a' aur a hamesha ek hi vertical **projector** par. Point HP par → a' XY par; VP par → a XY par.

3. Projection of Straight Lines

Notation: true length **TL (L)**, true inclination with HP = **θ**, with VP = **φ**; FV ka XY se angle = **α** (apparent), TV ka = **β**.

Position	FV	TV
∥ to both HP & VP	TL, ∥ XY	TL, ∥ XY
⊥ to HP (∥ VP)	TL, ⊥ XY	Point
⊥ to VP (∥ HP)	Point	TL, ⊥ XY
Inclined θ to HP, ∥ VP	TL at θ	L cos θ, ∥ XY
Inclined φ to VP, ∥ HP	L cos φ, ∥ XY	TL at φ
Inclined to both (oblique)	L cos φ at α (α > θ)	L cos θ at β (β > φ)

KEY RELATIONS θ + φ ≤ 90° (= 90° → line profile plane mein, FV & TV dono ⊥ XY). Difference of distances of ends from HP = L sin θ; from VP = L sin φ.

True length & true inclinations (line inclined to both) – rotating line method

1. Given FV a'b' aur TV ab draw karo (end projectors).
2. TV ab ko a centre lekar ghumao jab tak ∥ XY na ho (ab_1); b_1 se vertical projector, b' ka horizontal locus se mile → b_1'. **$a'b_1'$ = TL**, aur XY se angle = **θ**.
3. Isi tarah FV a'b' ko XY ke ∥ ghumao ($a'b_2'$), project down to b's locus → ab_2 = TL aur angle = **φ**.
4. Reverse problem (L, θ, φ given): pehle line ko θ par (∥ VP) aur φ par (∥ HP) draw karke lengths L cos θ, L cos φ nikaalo, phir loci ke beech arcs se final views.

Other methods: **trapezoid method** (TV ke ends par ⊥ heights = FV ki heights → TL) aur **auxiliary plane method** (X_1Y_1 ∥ to TV lekar auxiliary FV = TL).

Traces of a line

DEFINITION Line (ya extended line) jahan HP ko mile = **Horizontal Trace (HT)**; jahan VP ko mile = **Vertical Trace (VT)**. Line ∥ HP → HT nahi; ∥ VP → VT nahi; ∥ both → koi trace nahi.

1. **HT:** FV a'b' ko extend karo jab tak XY ko h' par mile; h' se vertical projector draw karo jo TV (extended) ko h par mile → **h = HT**.
2. **VT:** TV ab ko extend karo jab tak XY ko v par mile; v se vertical projector FV (extended) ko v' par mile → **v' = VT**.

4. Projection of Planes & Traces

Planes (laminae): triangle, square, rectangle, pentagon, hexagon, circle, semicircle. **Types:** (i) **perpendicular planes** – ⊥ to both (profile), ⊥ one & ∥ other, ⊥ one & inclined to other; (ii) **oblique plane** – inclined to both HP & VP.

Traces of plane: plane (extended) jis line par HP ko kaate = **HT**, VP ko kaate = **VT**. Plane ∥ HP → sirf VT (∥ XY); ⊥ HP & inclined VP → HT at φ, VT ⊥ XY; inclined HP & ⊥ VP → VT at θ, HT ⊥ XY. Oblique plane ke HT & VT XY par ek point par milte hain.

Plane position	True shape in	Other view
∥ HP (⊥ VP)	TV	FV = line ∥ XY
∥ VP (⊥ HP)	FV	TV = line ∥ XY
Inclined θ to HP, ⊥ VP	– (TV reduced)	FV = line at θ
Inclined φ to VP, ⊥ HP	– (FV reduced)	TV = line at φ

Plane inclined to both HP & VP – change of position (3 stages)

1. **Stage 1:** plane ko ∥ HP maan lo (agar θ given hai HP se) – TV true shape, FV line on XY. Given edge/corner ki condition ke hisaab se rakho (e.g. one side ⊥ VP).
2. **Stage 2:** FV line ko XY se θ par tilt karo; points project karke naya TV (reduced) banao.
3. **Stage 3:** stage-2 ke TV ko φ (ya β) se rotate karo jaisa given hai; FV ke points stage-2 FV ke horizontal loci aur naye TV ke projectors ke intersection se lo.

COMMON MISTAKE Edge/diagonal "inclined φ to VP" (true angle) aur "TV makes β with XY" (apparent angle) ko same mat samjho. φ ho to stage 3 mein TL-locus method lagana padta hai.

Auxiliary planes

Auxiliary Inclined Plane (AIP): ⊥ VP, inclined to HP. **Auxiliary Vertical Plane (AVP)**: ⊥ HP, inclined to VP. Inclined surface ka true shape dekhne ke liye naya reference line **X_1Y_1** us surface ke edge-view ke **parallel** lete hain; projectors X_1Y_1 ke ⊥ aur distances pichle (dusre) view se X_1Y_1 tak transfer karte hain → **auxiliary view** (true shape). Isse plane / line ki TL & true inclination bhi milti hai.

UNIT III Solids, Sections & Intersection of Surfaces

1. Types of Solids

Group	Examples & key points
Polyhedra (plane faces)	**Regular polyhedra**: tetrahedron (4 equilateral Δ), cube/hexahedron (6 squares), octahedron (8 Δ), dodecahedron (12 pentagons), icosahedron (20 Δ). **Prisms** (two equal parallel bases + rectangular faces) and **pyramids** (base polygon + triangular faces meeting at apex) – named by base: triangular, square, pentagonal, hexagonal.
Solids of revolution	**Cylinder** (rectangle rotated about a side), **cone** (right Δ about one side), **sphere** (semicircle about diameter).
Modified	**Frustum** (pyramid/cone ka part base ke ∥ plane se kaatne par), **truncated** (inclined plane se kaate), right vs oblique solids (axis ⊥ base or not).

2. Projection of Solids (first angle)

Position of axis	Which view first?
⊥ HP	TV first (true shape of base), then FV
⊥ VP	FV first (true base), then TV
∥ to both HP & VP	Side view first (true base), then FV & TV
Inclined θ to HP, ∥ VP	Stage 1: axis ⊥ HP → Stage 2: tilt FV by θ
Inclined φ to VP, ∥ HP	Stage 1: axis ⊥ VP → Stage 2: tilt TV by φ
Inclined to both HP & VP	3 stages (below)

Axis inclined to one plane, parallel to other – steps (e.g. cone, axis 30° to HP, ∥ VP)

1. **Stage 1:** solid ko simple position mein lo – axis ⊥ HP, base HP par. TV = circle (12 generators ke points), FV = triangle.
2. **Stage 2:** FV ko as it is copy karke is tarah tilt karo ki axis HP se 30° (ya base 60°) par ho; condition dhyan do – "resting on a point of base", "on a generator", "on an edge".
3. Naye FV ke har point se vertical projectors, aur stage-1 TV ke points se horizontal lines → intersections = naya TV. Visible/hidden edges check karke join karo.

Cylinder (generators ∥ axis, base ellipse dikhega), pyramid/prism (corners/edges number karke) sab mein same procedure. Polyhedra (tetrahedron, cube) bhi isi tarah.

Axis inclined to both HP & VP

1. Stage 1 & 2 upar jaise (axis θ to HP, ∥ VP).
2. **Stage 3:** stage-2 ka TV ko copy karke rotate karo taaki axis ka TV XY se **β** par ho (agar φ given ho to β pehle TL method se nikaalo). Naye TV ke points se projectors upar, stage-2 FV ke horizontal loci se mila kar final FV.

YAAD RAKHO Visibility rule: FV mein jo edges observer ke paas (TV mein XY se door) hain wo visible; TV mein jo edges upar hain (FV mein XY se door) wo visible. Outer boundary hamesha visible hoti hai.

3. Section of Solids

DEFINITION Solid ko imaginary **section (cutting) plane** se kaat kar andar ka shape dikhana. Kata hua surface = **section**, uska **true shape** tab milta hai jab use section plane ke ∥ plane par project karein. Cut surface par 45° thin **hatching** lines (2–3 mm spacing).

Section plane	Seen in	True shape kaise
∥ HP (⊥ VP)	FV = line ∥ XY	TV itself = true shape
∥ VP (⊥ HP)	TV = line ∥ XY	FV itself = true shape
Inclined to HP, ⊥ VP (AIP)	FV = line (VT) at θ	Auxiliary TV on X_1Y_1 ∥ VT
Inclined to VP, ⊥ HP (AVP)	TV = line (HT) at φ	Auxiliary FV on X_1Y_1 ∥ HT

Steps – section plane inclined to HP (e.g. hexagonal pyramid cut by AIP)

1. Solid ke FV & TV draw karo, sab corners name karo.
2. FV mein given angle/position par cutting plane (VT) draw karo; jahan ye edges/generators ko kaate wo points 1′, 2′, 3′ … mark karo.
3. Points ko neeche TV ke corresponding edges par project karo; join karke **sectional TV** hatch karo (cut part hata kar).
4. **True shape:** VT ke ∥ X_1Y_1 lo; 1′, 2′ … se ⊥ projectors, aur TV mein XY se distances X_1Y_1 se mark karo → join & hatch.

Cone ke sections: plane through apex → triangle; ∥ base → circle; inclined, all generators cut → ellipse; ∥ to a generator → parabola; ∥ axis (or angle with base > generator angle) → hyperbola. Cylinder: inclined plane → ellipse, ∥ axis → rectangle.

COMMON MISTAKE Cone/cylinder par points generators se project karo; agar point FV mein extreme generator par na ho to **horizontal circle method** (point se ∥ base circle) use karo, warna TV galat aata hai.

4. Intersection (Interpenetration) of Surfaces

Jab ek solid doosre mein ghusta hai to dono surfaces ki common line = **curve / line of intersection**. Use: pipe joints, ducts, boiler, chimney, machine castings.

(a) Line method: ek solid ki surface par kai lines (edges / generators) lo; har line doosre solid ki surface ko jahan pierce kare wo point dhoondo (us view se jahan doosra solid edge-view / circle mein dikhe). Prism–prism, cylinder–cylinder, prism–cylinder ke liye easy.

(b) Cutting plane method: dono solids ko kai parallel cutting planes se kaato (preferably jo dono par simple sections de – lines ya circles). Har plane par dono sections ke intersection points = curve ke points. Cone–cylinder, cone–prism ke liye best (horizontal planes → cone par circle).

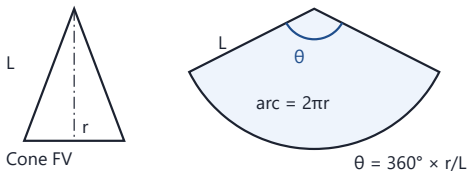
Case	Result / tip
Prism – prism	Intersection straight line segments; edge points line method se, phir visibility se join
Cylinder – cylinder (axes ⊥, intersecting)	Curve; equal diameters → FV mein straight lines (45° V shape)
Cylinder – prism	Prism edges cylinder ko kahan kaatte hain (TV circle se), phir intermediate lines
Cone – cylinder / cone – prism	Cutting plane method (horizontal planes): cone section = circle, cylinder section = rectangle lines

EXAM TIP Pehle "critical points" (extreme top/bottom/front/back aur outlines par) zaroor locate karo; curve unhi se visible/hidden part mein divide hota hai.

UNIT IV Development of Surfaces & Isometric Projection

1. Development of Surfaces

DEFINITION Solid ki saari surfaces ko ek plane par khol kar (unfold) true size mein dikhana = **development**. Sheet-metal work (buckets, funnels, ducts, hoppers, chimneys) mein pattern banane ke liye. Development mein har line **true length** mein hoti hai. Sirf **developable** surfaces (prism, cylinder, pyramid, cone) exactly khulte hain; sphere approximate.



Cone ki lateral surface ka development = sector (radius L, arc length = base circumference)

Method	Used for	Idea
Parallel line	Prisms, cylinders	Lateral edges/generators parallel; stretch-out line = perimeter, verticals par heights transfer
Radial line	Pyramids, cones	Edges/generators apex se radiate; slant edge/height ko radius lekar arc
Triangulation	Transition pieces, oblique cones	Surface ko chhote triangles mein todo, har triangle ka TL se true shape bana kar sequence mein lagao
Approximate	Sphere (zone / lune method)	Non-developable; segments approximate

FORMULA Cylinder: rectangle **πD × H**. Cone (base radius r, slant height L = √(r^2 + h^2)): sector of radius L, **θ = 360° × r / L**. Cube: 6 equal squares (cross / T shape). Prism: rectangles side by side, total length = perimeter of base. Pyramid: n triangles with radius = true slant edge length.

Steps – development of lateral surface of a cut (truncated) cone

1. FV & TV draw karo; TV circle ko 12 parts mein bato, generators o'1', o'2' … FV mein draw karo.
2. Radius = L (slant height) aur angle θ = 360 r/L se sector banao; sector ko 12 parts mein bato, generators O–1, O–2 … draw karo.
3. Cutting plane jahan generators ko kaate (FV mein), un points ko **true length** par laane ke liye horizontal line extreme generator (o'a', jo TL hai) tak le jao; O se wo TL distance development ke corresponding generator par mark karo.
4. Points ko smooth curve se join karo; jo part hataya gaya use thin/dashed dikhao. Prism/cylinder ke parts mein heights seedhe verticals par transfer hoti hain.

Pyramid development: slant edge ka TL nikaalo (edge ko FV mein extreme edge par ghuma kar); radius = TL se arc, uspar base side length n baar cut karo, apex se join. Cut pyramid mein cut points ko true length ke liye extreme (TL) edge par horizontal shift karo.

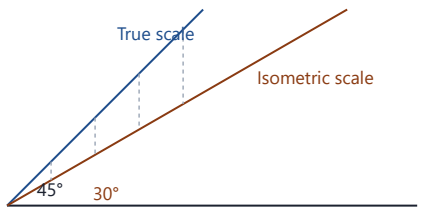
Transition pieces

Do alag shapes ya sizes ke openings ko jodne wala piece – e.g. **square-to-round**, round-to-rectangle (ducts, hoppers, chimney). Development triangulation se: round end ko 12 parts mein, har corner of square ko nearest circle points se join karke triangles + flat triangles bante hain; har line ka TL (TL diagram: horizontal = TV length, vertical = height) nikaal kar triangles ek-ek karke lagao.

2. Isometric Projection

DEFINITION Pictorial orthographic projection jisme cube ka body diagonal VP ke ⊥ hota hai, isliye teeno mutually perpendicular edges VP se **equally inclined** hote hain aur equally foreshortened. FV mein ek hi view mein teeno faces dikhti hain.

- **Isometric axes:** teen lines aapas mein **120°** par – ek vertical, do horizontal se **30°** par.
- **Isometric lines:** iso axes ke parallel lines (inpe hi measurement hoti hai). **Non-isometric lines:** parallel nahi (inclined edges) – unke end points coordinates se locate karke join karo (box / offset method).
- **Isometric planes:** do iso axes se bane plane – top (horizontal), left & right vertical faces.



Isometric scale: 45° line par true lengths, verticals se 30° line par project

ISOMETRIC SCALE

$$\text{Iso length} = \frac{\cos 45^\circ}{\cos 30^\circ} \times \text{TL} = \sqrt{(2/3)} \times \text{TL} \approx 0.816\ \text{TL}$$

Construction: horizontal line se 45° aur 30° lines draw karo; 45° line par true divisions (mm/cm) mark karo; har point se vertical drop karo 30° line tak → iso divisions.

Isometric VIEW (drawing)	Isometric PROJECTION
Iso lines par **true lengths** (ordinary scale)	Iso lines par **isometric lengths** (0.816 × true, iso scale)
Bada dikhta hai (≈ 22.5% larger)	Actual projection – exact size
Sphere: circle of radius ≈ 1.22 R	Sphere: circle of **true radius R**

Isometric of plane figures & solids

- **Rectangle/square/polygon:** enclosing rectangle (box) iso axes par banao, corners offsets se locate karo. Hexagon/pentagon ke liye box method.
- **Circle → ellipse** by **four-centre method**: circle ke around square ka isometric (rhombus) banao; obtuse-angle corners se opposite sides ke mid-points tak lines → do centres yahi corners, aur in lines ke intersection do aur centres; char arcs se ellipse. Ya **offset / co-ordinate method** (points plotting).
- **Cylinder:** top & bottom ellipses (four-centre), common tangents (extreme lines) se join; hidden half bottom ellipse dashed/omit.

- **Cone:** base ellipse + apex (axis par height mark) se ellipse ko tangent lines.
- **Pyramid:** base iso mein, centre se axis ∥ vertical iso axis par height, apex ko base corners se join (slant edges non-isometric).
- **Prism:** base iso mein draw karke har corner se axis ki direction mein length ∥ lines, top face join.
- **Sphere:** isometric projection = circle of radius R. Combination (e.g. sphere on cube/frustum): sphere ka contact point aur centre **iso scale** se locate karo (centre contact point se iso R upar), phir true R se circle.
- **Combination of solids:** neeche wale solid se shuru karo, upar wale ka base centre axis par locate karke draw karo; hidden lines generally nahi dikhate.

COMMON MISTAKE Non-isometric line (jaise pyramid ka slant edge, ya inclined face) par direct length mat napo – sirf iso lines par measure hota hai. Isometric drawing mein angle bhi true nahi dikhte.

UNIT V Computer Aided Drawing & Computer Graphics

1. CAD, CAED & CAE

Term	Meaning
CAD – Computer Aided Design / Drafting	Computer ka use design create, modify, analyse aur optimise karne ke liye; 2D drafting + 3D modelling (AutoCAD, SolidWorks, CATIA, Creo).
CAED – Computer Aided Engineering Design (/ Drafting)	CAD ka engineering design wala part: geometric modelling + engineering calculations + automated drafting, design ko documentation tak le jana.
CAE – Computer Aided Engineering	Design ka computer par analysis/simulation – FEA (stress), CFD (flow), kinematics, thermal (ANSYS, Abaqus) – prototype banane se pehle.

Design process mein CAD ka role

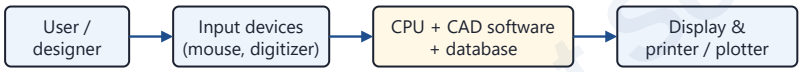
Design cycle: recognition of need → definition → synthesis → analysis & optimisation → evaluation → presentation. CAD mein: **geometric modelling** (synthesis), **engineering analysis** (CAE), **design review & evaluation** (interference check, animation), **automated drafting** (drawings, BOM).

2. Elements & Essential Requirements of CAD

Elements (components) of a CAD system:

- **Hardware** – CPU/workstation, memory, graphics card, input devices, display, output (printer/plotter), storage.
- **Software** – OS, graphics package (2D/3D), application & analysis software, database.
- **User / designer** – jo system ko operate kare (human creativity).
- Network & data-exchange (IGES, STEP, DXF).

Essential requirements: fast processor & sufficient RAM, high-resolution graphics display, accurate input devices, large storage, reliable software with standard drafting features (layers, snaps, dimensioning, libraries), good user interface, data security/back-up, trained operator, hard-copy output devices.



Block diagram of a CAD system (hardware + software + user)

Integrated CAD / CAM

CAM = Computer Aided Manufacturing (CNC part programming, process planning, robots, inspection). Integrated CAD/CAM mein ek hi **common database** design se manufacturing tak use hota hai – CAD model se seedha CNC tool-path (G-code) bante hain. Pura plant (design, planning, production, QC, business) jud jaye to **CIM** (Computer Integrated Manufacturing).

Necessity, importance & applications

Necessity / importance (advantages): productivity badhti hai, drafting time kam; high accuracy; easy modification & reuse (copy, mirror, array); better visualisation (3D, shading, animation); standardisation & libraries; analysis possible; easy storage, retrieval, sharing; less paper; lower cost & time-to-market; CAM se direct link.

Applications: mechanical (machine parts, assembly, CNC), civil & architecture (building plans, structures, GIS), electrical/electronics (PCB, circuits, VLSI), automobile & aerospace (body design, simulation), textile & fashion, ship building, plant layout & piping, animation & medical imaging.

YAAD RAKHO CAD systems ke types: mainframe-based, minicomputer-based, **workstation-based** aur **PC-based**. Modelling ke types: **2D drafting**, 3D **wireframe** (sirf edges), **surface** model (skin) aur **solid** model (mass, volume, interference check possible – sabse complete).

3. Computer Graphics – Input Devices

Device	Working / use
Cursor control devices	Screen par cursor move karke point/command select: **mouse** (optical/mechanical), **trackball**, **joystick**, **thumb wheels**, **light pen** (photocell screen se light sense kare), cursor keys.
Digitizer (graphic tablet)	Flat surface + **stylus / puck** (cross-hair); tablet ke andar wire grid electromagnetic signal se position (x, y) coordinates computer ko deta hai. Existing paper drawings ko CAD mein trace karne ke liye; high accuracy.
Keyboard terminals	Alphanumeric keys (text, coordinates, commands), function keys (programmable commands), cursor keys, numeric keypad.
Image scanner	Drawing/photo ko light se scan karke CCD sensors se digital raster image banata hai; raster-to-vector conversion software se CAD entity bana sakte hain.
Speech (voice) control	Microphone + voice recognition system; pre-recorded words ko pattern-match karke commands deta hai – hands free operation.
Touch panels	Screen ko ungli se chhoo kar select: **optical** (IR LEDs & detectors grid), **electrical/resistive** (do conductive sheets, pressure se contact), **capacitive**, **acoustic** (surface waves).

4. Display Devices

Cathode Ray Tube (CRT)

Parts: electron gun (heated cathode + control grid), focusing system, deflection system (electrostatic/magnetic plates – horizontal & vertical), phosphor-coated screen. Electron beam phosphor ko hit kare to light emit hoti hai; light jaldi decay hoti hai isliye picture ko baar-baar **refresh** karna padta hai (refresh CRT). **Persistence** = light ke 1/10 tak girne ka time; **resolution** = max points/pixels display; **aspect ratio** = vertical : horizontal points.

Random scan (vector / calligraphic)	Raster scan
Beam sirf un points par jaata hai jahan line draw karni hai	Beam pure screen ko row-by-row (left→right, top→bottom) sweep karta hai
Picture = line-drawing commands in **display file / refresh buffer**	Picture = intensity values of all **pixels** in **frame buffer** (bit-map)
Refresh 30–60 times/s; smooth, high-resolution lines	Refresh 60–80 frames/s (interlaced possible); lines staircase (**jaggies**/aliasing)
Realistic shaded/filled images mushkil; costly	Shading, colour, filled areas easy; cheap; TV technology
Use: line drawings, CAD wireframes (old)	Use: almost all modern monitors

CRT monitors (colour): (1) **Beam penetration** – red & green phosphor layers, beam speed se colour (only 4 colours, random scan). (2) **Shadow mask** – 3 electron guns (R, G, B), phosphor dots triad, shadow mask ke holes se har beam apne colour dot par – millions of colours (raster displays).

DVST (Direct View Storage Tube): picture **storage grid** par charge distribution ke roop mein store; **primary (writing) gun** pattern banata hai, **flood gun** continuously screen ko illuminate rakhta hai → **refresh ki zaroorat nahi**, flicker-free, complex drawings high resolution par. Drawbacks: selective erase nahi (poora screen erase karna padta hai), colour nahi, low contrast, animation possible nahi.

Flat panel displays: CRT se kam volume, weight & power. **Emissive** (electrical energy → light): plasma panel (gas-discharge, neon cells), thin-film electroluminescent, LED. **Non-emissive**: **LCD** – liquid crystal light ko polarise karke pass/block karta hai (backlight chahiye); passive & active-matrix (TFT). Laptops, tablets, modern monitors.

5. Hard Copy Devices – Printers & Plotters

Device	Types & working
Impact printers	Ribbon par hammer/pin se strike: **dot-matrix** (9/24 pins), daisy wheel, line/drum printer. Noisy, low quality, carbon copies possible.
Non-impact printers	**Inkjet** (ink droplets nozzles se), **laser** (laser beam drum par electrostatic image, toner, heat fusing – high quality & fast), thermal, electrostatic.
Pen plotters	Vector type, pen se continuous lines – large drawings (A0). **Flatbed** (paper fixed, pen X-Y move) aur **drum** (paper drum par ek axis mein move, pen doosre axis mein). Multiple coloured pens.
Electrostatic / inkjet plotters	Raster type; paper par charged dots + toner (electrostatic) ya ink droplets (inkjet plotter) – fast, large size, colour; aaj kal mostly inkjet wide-format plotters.

EXAM TIP Random vs raster scan, DVST ke advantages/disadvantages, aur digitizer ka working – ye 5–7 marks ke favourite questions hain. Table form mein likho aur CRT ka labelled block diagram (gun → focusing → deflection → phosphor screen) banao.